Virtual Environment for Rehabilitation of Upper Distal Limb Using a Haptic Device with Adaptive Impedance Control and Neural Compensation: A Preliminary Proposal

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Abstract

This research presents a preliminary proposal for a rehabilitation exercise aimed at patients with muscle weakness in the distal upper limb. A virtual environment was developed, where the user engages in a rehabilitation activity focused on rehabilitating the pinch grip. The goal is to strengthen the patient's grasp and reduce muscle weakness. The virtual environment was designed as a video game in order to generate greater interest and encourage patients to adhere to their rehabilitation activities. This virtual game utilizes the haptic device Novint Falcon for the interaction with the environment. This preliminary work implements an impedance control with neural compensation; the control strategy produces signals to adapt the force exerted by the patient, with the goal that the device can give a force of the same magnitude but in the opposite direction. Consequently, regardless of the patient's initial strength, the device will always deliver an assistive force to guide the patient along a desired trajectory. Initial experimental results with the proposed virtual-haptic rehabilitation system are presented, indicating the feasibility of the approach; however, further studies are required to validate its clinical effectiveness.

System Architecture

The proposed setup consists of three main modules:

- Virtual Environment: Developed in C++ using CHAI3D and SFML, the environment provides an interactive game where the user performs a pinch movement to control a virtual object.
- Haptic Device: A modified Novint Falcon equipped with 3D-printed end-effectors designed to simulate the pinch motion comfortably and precisely.
- Control System: The adaptive impedance controller ensures stable and natural force interaction, while a neural network compensates for dynamic uncertainties.

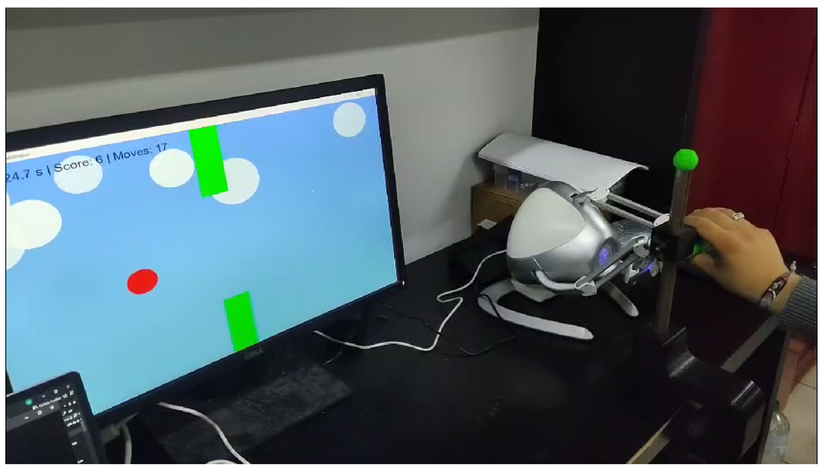


Figure 1. Rehabilitation session of a participant

Adaptive Impedance Control with Neural Compensation

The proposed control strategy allows the haptic device to interact safely and intuitively with the user by adjusting the resistance or assistance according to the user's effort.

This behavior is achieved by combining an impedance controller with a neural compensation mechanism. The impedance control defines a virtual relationship between the force applied by the device and its motion, similar to a spring–damper system:

$$M_a\ddot{\mathbf{x}} + B_a\dot{\mathbf{x}} + D_a x = \mathbf{f} \tag{1}$$

where M_a is the virtual inertia, B_a is the damping coefficient, and D_a is the stiffness coefficient. These parameters determine how "soft" or "stiff" the virtual environment feels to the user. The last equation can be rewritten in a matrix form, obtaining

$$\dot{\mathbf{x}} = Ax + B\mathbf{f} \tag{2}$$

where

$$A = \begin{bmatrix} 0 & 1 \\ -\frac{D_a}{M_a} & -\frac{B_a}{M_a} \end{bmatrix} \qquad B = \begin{bmatrix} 0 \\ \frac{1}{M_a} \end{bmatrix} \tag{3}$$

where $K_1, K_2 \in \mathbb{R}^{2\times 2}$ are positive defined diagonal matrices, and $\mathbf{f}$ is considered as an approximation of the force applied by the human $\mathbf{f_{human}}$, which is estimated as

$$\hat{\mathbf{f}}_{\mathbf{human}} = (\dot{\mathbf{x}} - \hat{A}x)\hat{B}^{-1} \tag{4}$$

Where $\hat{A}, \hat{B}$ represent the real-time estimates of the unknown dynamic parameters of the human–device interaction. An adaptive controller on the system can be applied, as expressed in Equation (2). For this, it is necessary to estimate the unknown terms of the model.

$$\dot{\hat{A}} = K_1\tilde{x}x_d^T, \qquad \dot{\hat{B}} = K_2\tilde{x}\mathbf{f} \tag{5}$$

where $K_1, K_2 \in \mathbb{R}^{2\times 2}$ are positive defined diagonal matrices, and $\mathbf{f}$ is considered as an approximation of the force applied by the human $\mathbf{f_{human}}$, which is estimated as

$$\hat{\mathbf{f}}_{\mathbf{human}} = (\dot{\mathbf{x}} - \hat{A}x)\hat{B}^{-1} \tag{6}$$

In this way, the control law can be determined by substituting $\dot{x}$ from Equation (6) into Equation (2), resulting in

$$\mathbf{u} = (-Ax + \hat{A}x + \hat{B}\hat{\mathbf{f}}_{\mathbf{human}})B^{-1} - \hat{\mathbf{f}}_{\mathbf{human}} \tag{7}$$

As can be seen, the matrix A and the vector B form the dynamic model of the system; however, these elements are unknown. For this research, it was decided to approach the issue using a neural network. For this case, their approximations are given by $A = \hat{M}_A, B = \hat{M}_B$,which are represented as outputs of a single-layer neural network.

$$\begin{aligned} \hat{M}_A(x,\dot{\mathbf{x}}) &= \hat{W}_A\sigma(x,\dot{\mathbf{x}}), \quad M_A(x,\dot{\mathbf{x}}) = W_A^*\sigma(x,\dot{\mathbf{x}}) + \phi(x,\dot{\mathbf{x}}) \\ \hat{M}_B(x,\dot{\mathbf{x}}) &= \hat{W}_B\sigma(x,\dot{\mathbf{x}}), \quad M_B(x,\dot{\mathbf{x}}) = W_B^*\sigma(x,\dot{\mathbf{x}}) + \phi(x,\dot{\mathbf{x}}) \end{aligned} \tag{8}$$

where W_A^*, W_B^* are unknown constant weights, $\hat{W}_A, \hat{W}_B$ are estimated weights, $\phi(x,\dot{\mathbf{x}})$ is a neural approximation error, and σ is a function of neuronal activation.

In this way, compensating the dynamics not modeled can be achieved, adding robustness to the control. In this way, the control law is rewritten as follows:

$$\mathbf{u} = (-\hat{M}_A x + \hat{A}x + \hat{B}\hat{\mathbf{f}}_{\mathbf{human}})\hat{M}_B^{-1} - \hat{\mathbf{f}}_{\mathbf{human}} \tag{9}$$

Results

During each trial, the participant interacted with the virtual environment by performing a pinch movement with the Novint Falcon. The goal was to move a small object along a fixed path and avoid obstacles displayed on the screen. The system recorded position and force data while the adaptive impedance controller continuously adjusted the assistance level.

Simulation results (MATLAB):
The controller maintained trajectory tracking with less than 0.06 MSE and 0.12 MAE under variable force conditions.

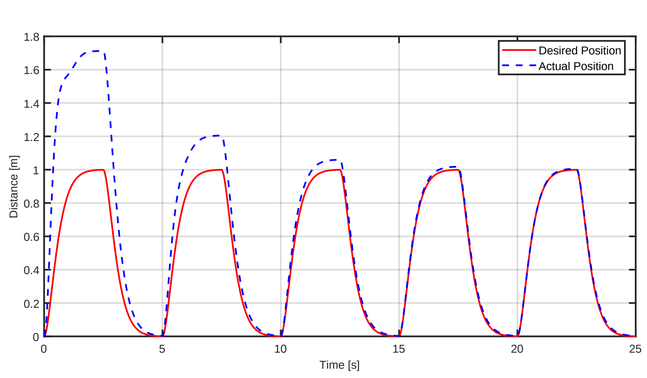


Figure 2. Position tracking comparison (simulation).

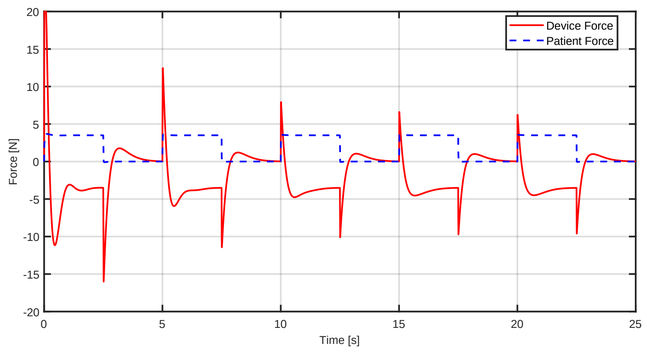


Figure 3. Forces applied by user and device (simulation).

Experimental results (10 healthy subjects):
Ten volunteers (aged 21–32) performed the virtual task for about 100 s per session. The adaptive controller increased assistance when effort was low and reduced it when excessive force was applied.

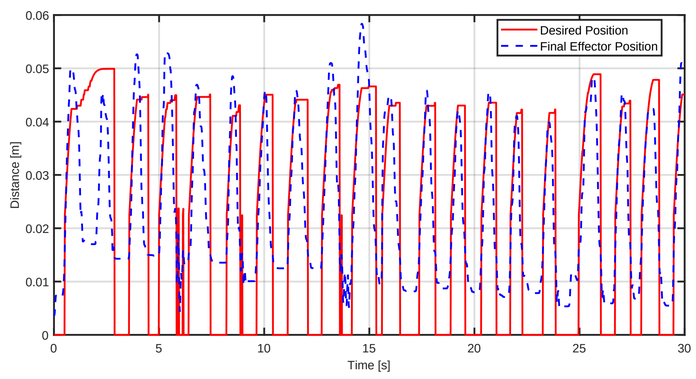


Figure 4. Position tracking comparison (experiment).

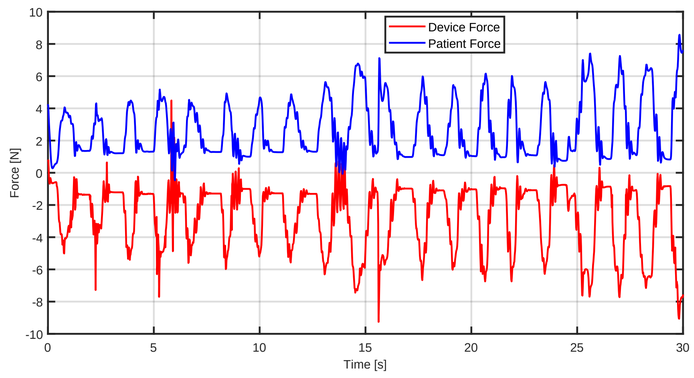


Figure 5. Forces applied by user and device (experiment).

Conclusions

The proposed system successfully integrates virtual reality, haptic feedback, and intelligent control to support personalized rehabilitation of the hand and fingers. By combining adaptive impedance control with neural compensation, the device can automatically adjust the level of force assistance or resistance according to the user's performance, providing a stable and natural interaction. Experimental results with healthy participants confirmed that the controller maintains trajectory tracking and adapts to varying user efforts in real time. These findings demonstrate the feasibility of implementing an adaptive, low-cost haptic rehabilitation platform capable of enhancing motivation and safety during training. Future studies will focus on testing the system with patients who present muscular weakness or post-stroke motor impairments, in order to evaluate its therapeutic effectiveness and long-term adaptability.

References

[1] Yahel Cortés-García, Yukio Rosales-Luengas, Saul J Rangel-Popoca, Sergio Salazar, Xiaoou Li, and Rogelio Lozano. Virtual environment for rehabilitation of upper distal limb using a haptic device with adaptive impedance control and neural compensation: A preliminary proposal. *Sensors*, 25(19):5964, 2025.